LF-FMR040YS0350U(S)

FMR*YS*U non-SELV 1 driver with 4-output current | Constant current - Non dimmable

Product family features

— Low THD≦15%@full load

— Rated input voltage: 100-240Vac

— Ta: -30℃~+60℃

— Ripple current<5%

— 5 years guarantee

Product family benefits

— Output current adjustable via DIP switch in 4-shifts

— Linear metal casing with 21mm housing height

— Long lifetime and high reliability

— Flicker free; non-SELV output

Typical applications

— For linear light, strip light and tri-proof light

— For office, commercial and decorative lighting

Product parameters

— Output current 200/250/300/350mA

— Output power 16-40.2W

— Input voltage 100–264Vac

— Output voltage 45-130Vdc

— Efficiency 91%

Electrical data

Input data

Rated supply voltage	100 ... 240 V
AC voltage range	100 ... 264 V
Mains frequency	50/60 Hz
Power factor	≥0.95[1)]
Efficiency in max. power	≧91% [2)]
THD	≦15%
Input current	0.50A Max
Inrush current	48A[3)]
Loading number on circuit breaker 10 A (B)	8
Loading number on circuit breaker 10 A (C)	13
Loading number on circuit breaker 16 A (B)	13
Loading number on circuit breaker 16 A (C)	22
Protective conductor current	≤3.5mA

Output data

Nominal output voltage	45 ... 130V [4)]
Nominal output current	200/250/300/350mA [5)]
Default output current	350mA
Current set	DIP Switch (please see the DIP switch definition)
Maximum output power	40.2W
Nominal output power	16 ... 40.2W [6)]
Output ripple current (100 Hz)	<5%
Flicker	Comply with IEEE Std 1789-2015
CIE SVM	≤0.4
IEC-Pst	≤1
Output current tolerance	±5%
Temperature tolerance	±10%
Starting time	<0.5S

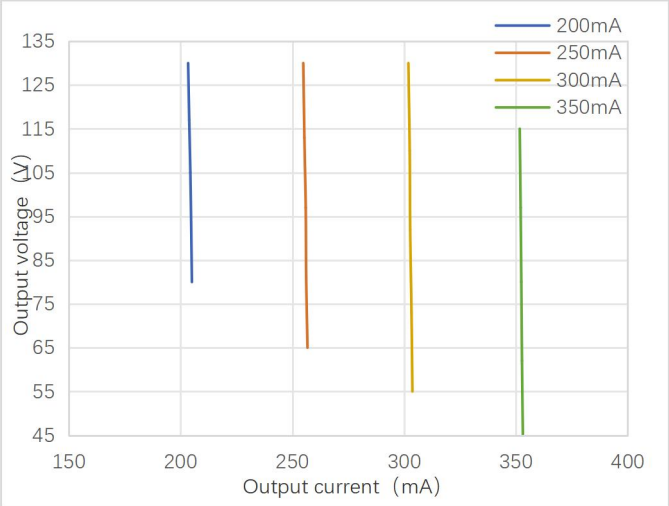
Safety

Withstanding voltage	I/P-PG O/P-PG: 1.5kV&5mA&60S
Surge capability (L-N)	2 kV
Surge capability (L/N--Ground)	2 kV
Insulation resistance	I/P-PG O/P-PG: ＞100MΩ@500VDC
Guarantee	5 years [7)]

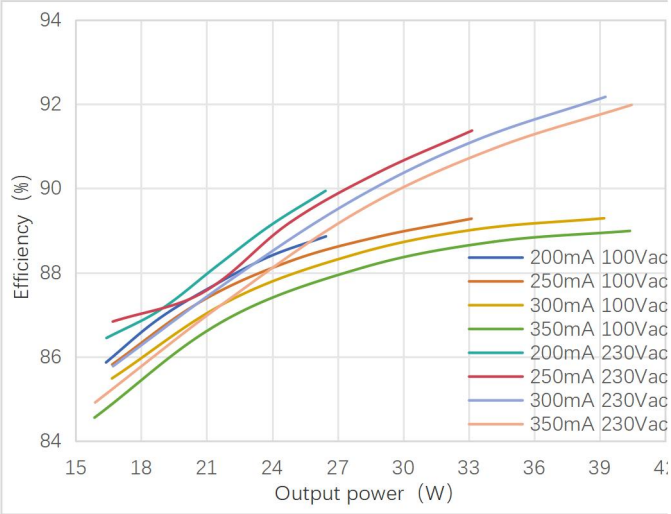
1) 0.95@300-350mA; 0.94@250mA; 0.93@200mA t = 220μs
2) 91%@300-350mA; 89%@250mA; 88%@200mA
3) t = 210μs
4) For the relationship between output voltage and current, refer to the operating window
5) Fixed current version optional
6) 50%-100% load of the product meets the requirements of single harmonics in European regulation EN61000-3-2
7) 5 years@100Vac@Tc≦85℃, 5 years@230Vac@Tc≦85℃

Characteristic diagram

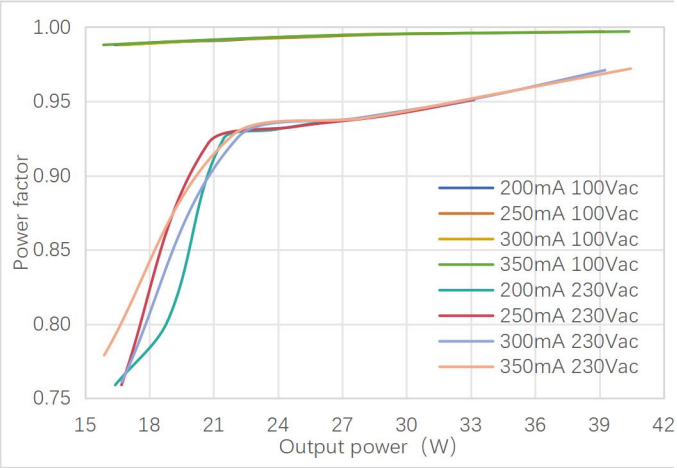
Operating Window



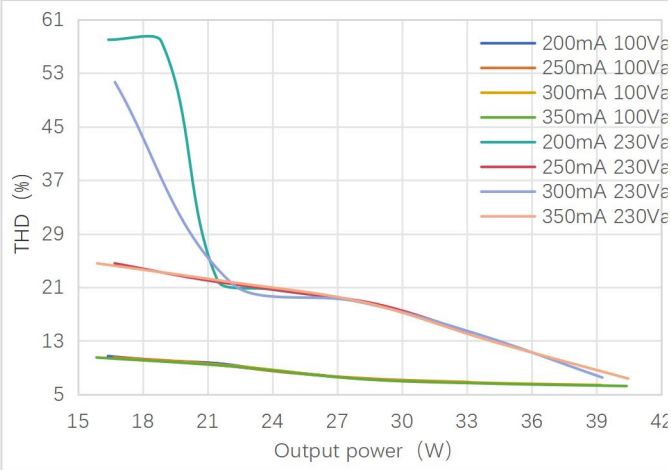
Typical Efficiency vs Load



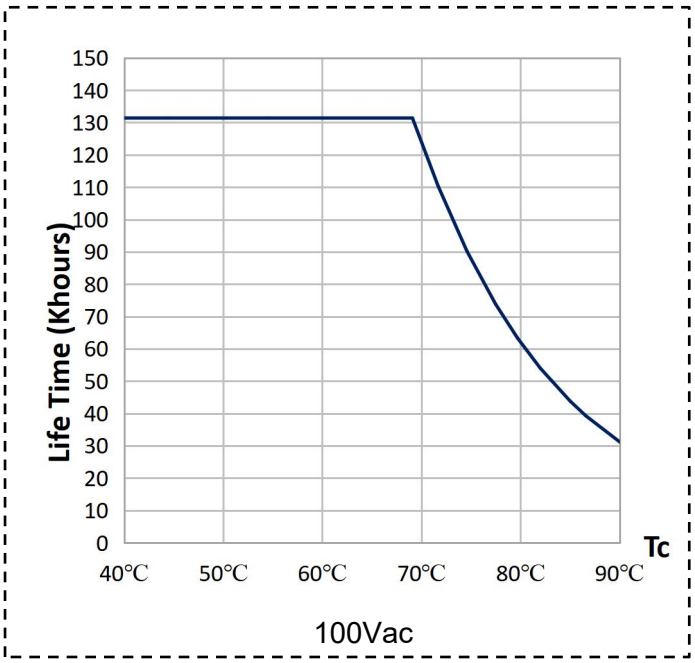
Typical Power Factor vs Load

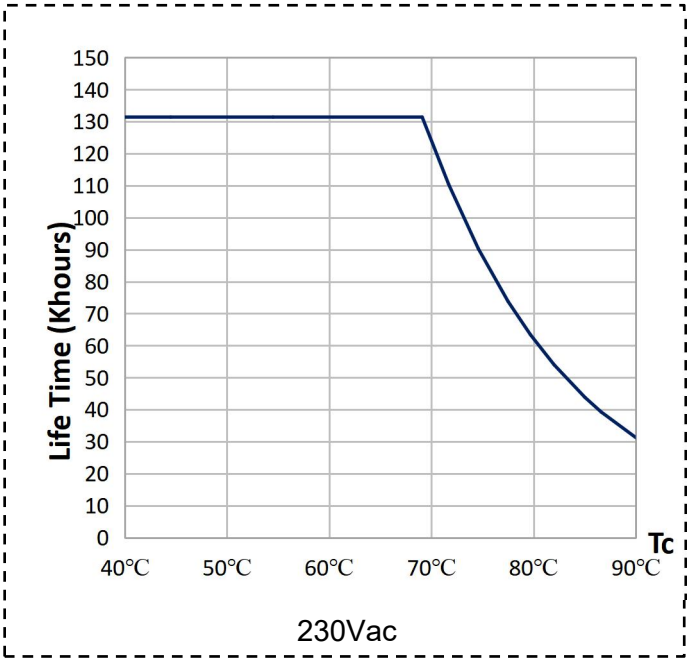


Typical THD vs Load

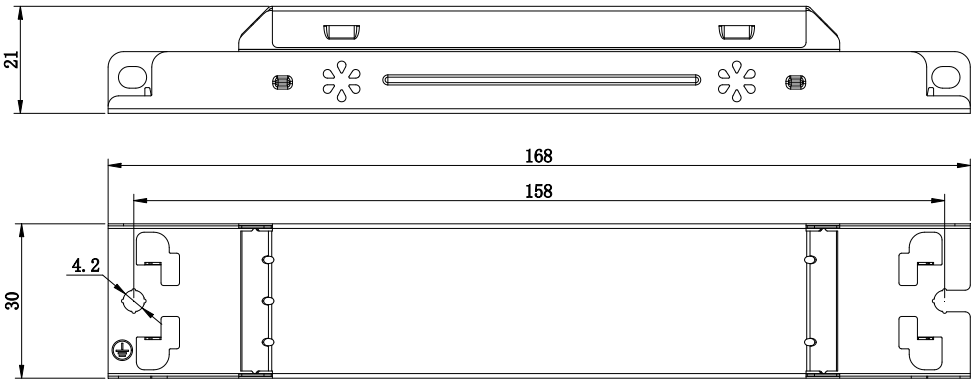


Lifespan





Dimensions



Mounting hole spacing length	158.0mm
Mounting hole diameter	4.2mm
Product weight	120.0g
Cable cross-section, input side	0.5 ... 1.5 mm²
Cable cross-section, output side	0.5 ... 1.5 mm²
Wire preparation length, input side	7 ... 8mm
Wire preparation length, output side	7 ... 8mm
Length	168.0mm
Width	30.0mm
Height	21.0mm

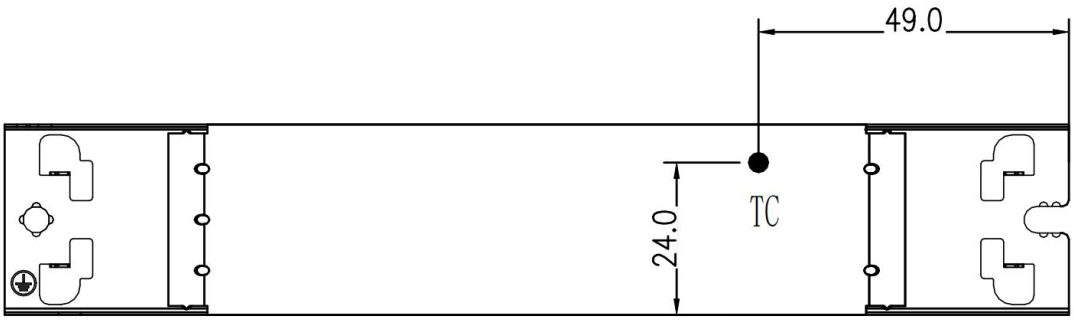
Colors & materials

Casing material	Color coated galvanized sheet
Casing color	White

Temperature & operating conditions

Ambient temperature range	-30℃ - +60℃
Maximum temperature at Tc test point	90℃
Temperature range at storage	-30℃ - +80℃ (6 months in Class I environment)
Humidity range at storage	20-95%RH (no condensation)
Humidity during operation	20-90%RH
RoHS	RoHS 2.0 (EU) 2015/863

Tc test point



Note: This diagram is a front view and Tc point is on the front side of the LED driver.

Product terminal

Input		Output	
AC-L	AC live wire input	**LED+**	Positive terminal output of LED driver
AC-N	AC neutral wire input	**LED-**	Negative terminal output of LED driver
⏚	Earth wire		

DIP switch terminal

Output current	Output voltage	DIP switch 1	DIP switch 2
200mA	80-130Vdc	-	-
250mA	65-130Vdc	-	ON
300mA	55-130Vdc	ON	-
* 350mA	45-115Vdc	ON	ON

Note: “-”: shift OFF. “*”: default current. DIP when power on is NOT allowed. Please disconnect the AC power before DIP.

Capabilities

Dimmable	-
Over heating protection	-
Overload protection	-
Short-circuit protection	Auto-recovery

No-load protection	<250V
Suitable for fixtures with prot. class	I
Control interface	-
Number of channels	1 channel

Programming

Programmer	-
DALI control software	-
APP	-

Certificates & standards

Approval marks – approval	CE, CB, ENEC
Standards	IEC/EN 61347-2-13, IEC/EN 61347-1, IEC/EN 62493 IEC/EN 62384
EMC	GB 17625.1-2022, GB/T 17743-2021 EN 55015, EN 61547, EN 61000-3-2,3
Type of protection	IP20

Logistical data

Product	Packaging unit (Pieces/Unit)	Dimensions (L*W*H)	Volume	Gross weight
LF-FMR040YS0350U(S)	84	385mm*285mm*210mm	23.04 dm³	10.76kg±5%

Test equipment & condition

Test Equipment	AC power source: CHROMA6530, digital power meter: CHROMA66205, oscilloscope: Tektronix DPO3014, DC electronic load: M9712B, LED board, constant temperature and humidity chamber, lightning surge generator: Everfine EMS61000-5B, rapid group pulse generator: Everfine EMS61000-4A, spectroanalyzer: KH3935, hi-pot tester: EEC SE7440, flicker tester (flicker-free coefficient test): Everfine LFA-3000, etc.

If there are no special remarks, the above parameters are tested at the ambient temperature of 25℃, humidity of 50%, full load and input voltage of 230Vac/50Hz.

Additional information

1. It is recommended that user install the over voltage protection, under voltage protection and surge protection devices in the power supply circuits of light fixtures to ensure electricity safety.

2. The LED driver used in combination with the end device is one of the accessories of the whole light fixture, and the EMC of the whole light fixture is not only susceptible to the driver itself, but to the LED light fixture and the whole light fixture's wiring. Thus, the manufacturer of LED light fixture should re-confirm the EMC of the whole light fixture before the whole light fixture is finished.

3. The test conditions of the circuit breaker configuration quantity are the same as those of the inrush current.

4. The PC cover, casing and end cap for assembling the LED driver in the light fixture must meet the fire rating of UL94-V0 or above.

5. It is well-advised that the withstanding voltage of LEDs and aluminum substrates >3kV.

6. It is recommended to install double-pole switch at AC input terminal. If user uses the single-pole switch, make sure to connect it to wire L (live wire), otherwise the afterglow of light fixture would be incurred after the AC is disconnected.

7. Because there is parasitic capacitance between LEDs and the PCBA, and the PCBA (the light fixture) is grounding, there will be a slight flicker at the moment of AC power on. It's normal for non-isolated products.

8. The light panel, fixed bracket and driver grounding should be secure.

Transportation & storage

Suitable transportation means: vehicles, boats and aeroplanes.

In transit, it is necessary to prepare awnings for rain or sun protection. Moreover, please keep civilized loading and unloading to prevent the vibration or impact of LED driver as much as possible.

The storage of LED driver shall conform to the standard of Class I environment. When using LED drivers which have been stored for more than 6 months, please re-test them firstly. Do not use them unless they are tested to be qualified.

Cautions

Please use Lifud LED driver according to its parameters in the specification, otherwise the LED driver may malfunction.

Using any incompatible light fixtures or those that have not been certified may cause fire, explosion or other risks.

Man-made damage is beyond the scope of Lifud warranty service.

Disclaimer

Subject to change without notice. Errors and omission excepted. Always make sure to use the most recent release.

Lifud Technology Co., Ltd. reserves the right to interpret any contents of this specification.